

Last Time

Testing Equality of Variances
The F-Distribution

Introduction to ANOVA

Logic of ANOVA
Partitioning the Variance

Today

Review

Sources of Variation

ANOVA Model

- Sum of Squares Within

- Sum of Squares Between

- The F-statistic

Example

- The Summary Table

Effect Size and Power

Introductory Remarks about ANOVA

- ▶ How do we compare several (more than two) populations?

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

Introductory Remarks about ANOVA

- ▶ How do we compare several (more than two) populations?

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

- ▶ Consider pairwise t -tests, i.e.

$$H_0 : \mu_1 = \mu_2$$

$$H_0 : \mu_1 = \mu_3$$

$$\vdots$$

$$H_0 : \mu_{k-1} = \mu_k$$

An example using a *nominal* independent variable

- ▶ Patients with a fear of heights are randomly assigned to one of five psychotherapy treatments:
 - ▶ Gestalt
 - ▶ Psychoanalysis
 - ▶ Rational Emotive
 - ▶ Behavior Modification
 - ▶ No Treatment (Control)
- ▶ Measure ratings of fear after treatment.

Back to the Null Hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

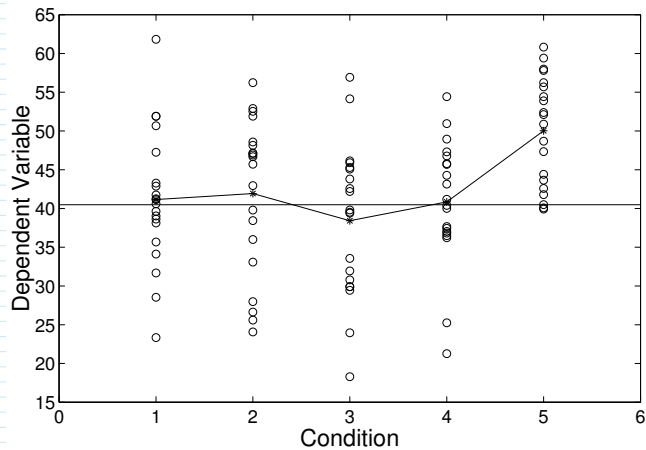
- ▶ This null hypothesis is sometimes called the *omnibus* hypothesis.
- ▶ The ANOVA performed to reject H_0 is sometimes called the *omnibus test* or *omnibus F*.

Back to the Null Hypothesis

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

- ▶ This null hypothesis is sometimes called the *omnibus* hypothesis.
- ▶ The ANOVA performed to reject H_0 is sometimes called the *omnibus test* or *omnibus F*.
- ▶ The alternative hypothesis is $H_1 : \text{not } H_0$, at least one of the means is different from the others.

Partitioning the variance



Partitioning the Sums of Squares SSX

Recall that

$$SSX = \sum_i (X_i - \bar{X})^2.$$

Let X_{ij} represent the score of individual i in condition j .
Over all groups,

$$SS_{\text{Total}} = \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2$$

Formal partitioning

$$\begin{aligned}SS_{\text{Total}} &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2 = \sum_{j=1}^k \sum_{i=1}^{N_j} [(X_{ij} - \bar{X}_j) + (\bar{X}_j - \bar{\bar{X}})]^2 \\&= \sum_{j=1}^k \sum_{i=1}^{N_j} [(X_{ij} - \bar{X}_j)^2 + 2(X_{ij} - \bar{X}_j)(\bar{X}_j - \bar{\bar{X}}) + (\bar{X}_j - \bar{\bar{X}})^2] \\&= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 + \sum_{j=1}^k \sum_{i=1}^{N_j} 2(X_{ij} - \bar{X}_j)(\bar{X}_j - \bar{\bar{X}}) + \sum_{j=1}^k \sum_{i=1}^{N_j} (\bar{X}_j - \bar{\bar{X}})^2 \\&= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 + \sum_{j=1}^k \sum_{i=1}^{N_j} (\bar{X}_j - \bar{\bar{X}})^2 \\&= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 + \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2 \\SS_{\text{Total}} &= SS_{\text{Within}} + SS_{\text{Between}}\end{aligned}$$

Sources of variation

- ▶ Within-groups variance reflects the deviation of any score X_{ij} from its group mean $\bar{X}_j$.
- ▶ Between-groups variance reflects the deviation of any mean $\bar{X}_j$ from the overall mean $\bar{\bar{X}}$.

Sources of variation

- ▶ Within-groups variance reflects the deviation of any score X_{ij} from its group mean $\bar{X}_j$.
- ▶ Between-groups variance reflects the deviation of any mean $\bar{X}_j$ from the overall mean $\bar{\bar{X}}$.
- ▶ Note to remember for later: Remember how s^2 is an estimate of σ^2 ? Within- and Between-groups variance are also estimates of the same population variance σ_e^2 if H_0 is true.

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ (violations of this assumption are not serious if N_{Total} is large)

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ (violations of this assumption are not serious if N_{Total} is large)
- ▶ Homoscedasticity: equal variances across groups
 - ▶ (violations are not serious if N_{Total} is large or if N_j s are equal across groups)

Assumptions of ANOVA

- ▶ The populations (X_j) are normally distributed
 - ▶ (violations of this assumption are not serious if N_{Total} is large)
- ▶ Homoscedasticity: equal variances across groups
 - ▶ (violations are not serious if N_{Total} is large or if N_j s are equal across groups)
- ▶ Each group is independent (otherwise we have a repeated measures design)

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

	Statistic	Parameter
Group mean	$\bar{X}_j$	μ_j
Grand mean	$\bar{\bar{X}}$	μ
Error	$X_{ij} - \bar{X}_j$	e_{ij}
Effect	$\bar{X}_j - \bar{\bar{X}}$	$\tau_j = \mu_j - \mu$
Error $e_{ij} \sim \mathcal{N}(0, \sigma_e^2)$		

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

► Model:

$$\begin{aligned} X_{ij} &= \mu_j + e_{ij} \\ &= \mu + (\mu_j - \mu) + e_{ij} \\ &= \mu + \tau_j + e_{ij} \end{aligned}$$

The ANOVA Model

Let X_{ij} be the measured variable for Individual i in Group $j = 1, \dots, k$.

► Model:

$$\begin{aligned}X_{ij} &= \mu_j + e_{ij} \\&= \mu + (\mu_j - \mu) + e_{ij} \\&= \mu + \tau_j + e_{ij}\end{aligned}$$

- If H_0 is true, $\mu_1 = \dots = \mu_k$, so $\mu_j = \mu$, the grand mean, for all j .
- If H_0 is true, $\tau_j = \mu_j - \mu = 0$.

What is SS_{Within} ?

$$SS_{\text{Within}} = \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 = \sum_{j=1}^k (N_j - 1) s_j^2$$

What is SS_{Within} ?

$$SS_{\text{Within}} = \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 = \sum_{j=1}^k (N_j - 1) s_j^2$$

If homoscedasticity is assumed, then SS_{Within} is the numerator of a *pooled variance* estimate of σ_e^2 :

$$\hat{\sigma}_e^2 = s_P^2 = \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{(N_1 - 1) + (N_2 - 1) + \cdots + (N_k - 1)}$$

What is SS_{Within} ?

$$SS_{\text{Within}} = \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 = \sum_{j=1}^k (N_j - 1) s_j^2$$

If homoscedasticity is assumed, then SS_{Within} is the numerator of a *pooled variance* estimate of σ_e^2 :

$$\hat{\sigma}_e^2 = s_P^2 = \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{(N_1 - 1) + (N_2 - 1) + \cdots + (N_k - 1)}$$

Note: $df = (N_1 - 1) + (N_2 - 1) + \cdots + (N_k - 1) = N_{\text{Total}} - k$

What is SS_{Between} ?

SS_{Between} is more difficult to interpret.

What is SS_{Between} ?

SS_{Between} is more difficult to interpret.

- If H_0 is *true*, $\sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2$ can *also* be used to estimate σ_e^2

What is SS_{Between} ?

SS_{Between} is more difficult to interpret.

- ▶ If H_0 is *true*, $\sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2$ can *also* be used to estimate σ_e^2
- ▶ If H_0 is *false*, SS_{Between} reflects σ_e^2 *plus* the effects of the treatment

What is SS_{Between} ?

SS_{Between} is more difficult to interpret.

- ▶ If H_0 is *true*, $\sum_{j=1}^k N_j(\bar{X}_j - \bar{\bar{X}})^2$ can *also* be used to estimate σ_e^2
- ▶ If H_0 is *false*, SS_{Between} reflects σ_e^2 *plus* the effects of the treatment
- ▶ Note that if we use $\sum_{j=1}^k N_j(\bar{X}_j - \bar{\bar{X}})^2$ to estimate σ_e^2 , it has $k - 1$ degrees of freedom.

The Rationale Behind ANOVA

- If H_0 is true, we have two estimates of σ_e^2 :

$$\frac{SS_{\text{Within}}}{N_{\text{Total}} - k} = \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{N_{\text{Total}} - k}$$
$$\frac{SS_{\text{Between}}}{k - 1} = \frac{\sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2}{k - 1}$$

The Rationale Behind ANOVA

- ▶ If H_0 is true, we have two estimates of σ_e^2 :

$$\frac{SS_{\text{Within}}}{N_{\text{Total}} - k} = \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{N_{\text{Total}} - k}$$
$$\frac{SS_{\text{Between}}}{k - 1} = \frac{\sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2}{k - 1}$$

- ▶ Call

$$\frac{SS_{\text{Within}}}{N_{\text{Total}} - k} = MS_W$$

Mean Square Within.

- ▶ Call

$$\frac{SS_{\text{Between}}}{k - 1} = MS_B$$

Mean Square Between.

Estimates of σ_e^2

$$MS_{\text{Within}} = \frac{1}{N_{\text{Total}} - k} \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 = \frac{\sum_{j=1}^k (N_j - 1) s_j^2}{N_{\text{Total}} - k}$$

$$MS_{\text{Between}} = \frac{1}{k - 1} \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2$$

$$E(MS_{\text{Within}}) = \sigma_e^2$$

$$E(MS_{\text{Between}}) = \sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k - 1}$$

Estimates of σ_e^2

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \end{aligned}$$

The F-Statistic

How do we test for equality of two variances?

The statistic

$$F = \frac{MS_B}{MS_W}$$

is distributed as an F statistic with $(k - 1, N_{\text{Total}} - k)$ degrees of freedom.

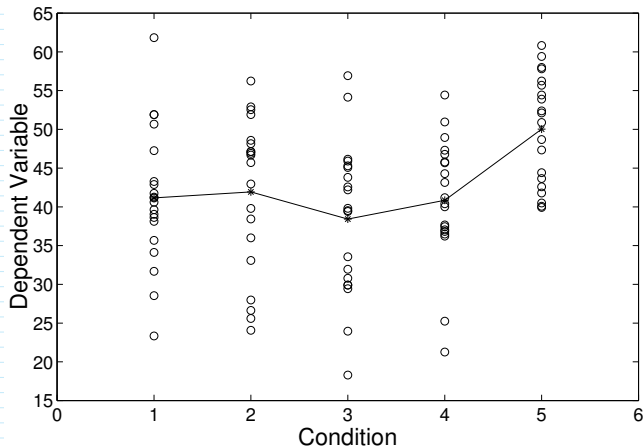
The omnibus F -test

If H_0 is false,

$$\begin{aligned} F &= \frac{MS_B}{MS_W} \\ &= \frac{\text{Estimate of } \sigma_e^2 + (\text{Treatment Effects})^2}{\text{Estimate of } \sigma_e^2} \end{aligned}$$

Example

Remember the therapy options for patients with fear of heights. After the treatment, each patient is taken to a building ledge and asked to rate his/her fear on a scale from 0-100.



Statistics

	1	2	3	4	5
$\bar{X}_j$	40.16	42.45	42.55	39.89	53.50
N_j	20	20	20	20	20

Statistics

	1	2	3	4	5
$\bar{X}_j$	40.16	42.45	42.55	39.89	53.50
N_j	20	20	20	20	20

Computing the Sums of Squares

$$\begin{aligned}SS_T &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2 \\ &= 12,525.03\end{aligned}$$

Computing the Sums of Squares

$$\begin{aligned}SS_T &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2 \\&= 12,525.03 \\SS_B &= \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2 \\&= 2,519.03\end{aligned}$$

Computing the Sums of Squares

$$\begin{aligned}SS_T &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2 \\ &= 12,525.03\end{aligned}$$

$$\begin{aligned}SS_B &= \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2 \\ &= 2,519.03\end{aligned}$$

$$\begin{aligned}SS_W &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 \\ &= 10,006.00\end{aligned}$$

Computing the Sums of Squares

$$\begin{aligned}SS_T &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{\bar{X}})^2 \\ &= 12,525.03\end{aligned}$$

$$\begin{aligned}SS_B &= \sum_{j=1}^k N_j (\bar{X}_j - \bar{\bar{X}})^2 \\ &= 2,519.03\end{aligned}$$

$$\begin{aligned}SS_W &= \sum_{j=1}^k \sum_{i=1}^{N_j} (X_{ij} - \bar{X}_j)^2 \\ &= 10,006.00 \\ &= SS_T - SS_B = 12,525.03 - 2,519.03\end{aligned}$$

The ANOVA Summary Table

Source	SS	df	MS	F
Between				
Within				
Total				

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03			
Within	10,006.00			
Total				

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03			
Within	10,006.00			
Total	12,525.03			

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4		
Within	10,006.00			
Total	12,525.03			

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	
Within	10,006.00			
Total	12,525.03			

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	
Within	10,006.00	95		
Total	12,525.03			

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	
Within	10,006.00	95		
Total	12,525.03	99		

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	
Within	10,006.00	95	105.33	
Total	12,525.03	99		

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Reject or fail to reject H_0 ?

The ANOVA Summary Table

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Reject or fail to reject H_0 ?

$$F_{.05}(4, 95) = 2.4675$$

$$p\text{-value} = 2.46 \times 10^{-4} < 0.05$$

In R

```
> # ANOVA in R
> group.factor <- as.factor(group)
> results <- aov(X~group.factor)
> summary(results)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
group.factor	4	2519	629.8	5.979	0.000246 ***
Residuals	95	10006	105.3		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Caution!

An ANOVA requires (REQUIRES!) that the predictor/independent variable be transformed to a factor. It must be categorical! (!!!)

```
> results <- aov(X~group)
> summary(results)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
group	1	1163	1163.4	10.04	0.00205	**
Residuals	98	11362	115.9			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Another Example

Participants rate guilt/innocence on a 10-point scale (1 = certain she is innocent, 10 = certain she is guilty)

Juror	Record	Group	
		Clean	Control
1	10	5	4
2	7	1	6
3	5	3	9
4	10	7	3
5	8	4	3

Another Example

Participants rate guilt/innocence on a 10-point scale (1 = certain she is innocent, 10 = certain she is guilty)

Juror	Record	Group		
		Clean	Control	
1	10	5	4	
2	7	1	6	
3	5	3	9	
4	10	7	3	
5	8	4	3	
$\bar{X}_j$	8	4	5	$\bar{\bar{X}}=5.67$

The ANOVA Summary Table

Source	SS	df	MS	F
Between				
Within				
Total				

Reject or fail to reject H_0 ?

ANOVA and Power

The same factors affect power in the ANOVA as in z and t tests:

- ▶ The effect sizes, the true differences between the means
- ▶ The variance of the population (σ_e^2)
- ▶ Numerator $df = k - 1$
- ▶ Denominator $df = N_{\text{Total}} - k$
- ▶ The α -level

ANOVA Effect Size

- ▶ Cohen's d is a distance measure.
- ▶ ANOVA relies on r measures.
- ▶ r measures are a function of the correlation between dependent and independent variables.

ANOVA Effect Size

- ▶ Cohen's d is a distance measure.
- ▶ ANOVA relies on r measures.
- ▶ r measures are a function of the correlation between dependent and independent variables.
- ▶ Two measures:
 - ▶ η^2
 - ▶ ω^2

$$\eta^2$$

$$\eta^2 = \frac{SS_{\text{Between}}}{SS_{\text{Total}}}$$

- ▶ Proportion of explained variance (R^2)
- ▶ Percent reduction in error

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

$$\eta^2 = \frac{2,519.03}{12,525.03} = 0.2011$$

ω^2

η^2 is biased: it overestimates the proportion of “explained” variance.

ω^2 is a correction to η^2 :

$$\omega^2 = \frac{SS_{\text{Between}} - (k - 1)MS_{\text{Within}}}{SS_{\text{Total}} + MS_{\text{Within}}}$$

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

Example: Fear of Heights

Source	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>
Between	2,519.03	4	629.76	5.98
Within	10,006.00	95	105.33	
Total	12,525.03	99		

$$\omega^2 = \frac{2,519.03 - (5 - 1)(105.33)}{12,525.03 + 105.33} = 0.1661$$

T-shirt Sizes

For both η^2 and ω^2 ,

Small	0.01
Medium	0.06
Large	0.14

Noncentrality Parameter

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \\ &= 1 + \frac{\sum_{j=1}^k N_j \tau_j^2 / (k-1)}{\sigma_e^2} \end{aligned}$$

Noncentrality Parameter

$$\begin{aligned} F &\approx \frac{E(MS_{\text{Between}})}{E(MS_{\text{Within}})} \\ &= \frac{\sigma_e^2 + \frac{\sum_{j=1}^k N_j \tau_j^2}{k-1}}{\sigma_e^2} \\ &= 1 + \frac{\sum_{j=1}^k N_j \tau_j^2 / (k-1)}{\sigma_e^2} \end{aligned}$$

$$\lambda = \frac{\sum_{j=1}^k N_j \tau_j^2 / (k-1)}{\sigma_e^2}$$

In R

```
> results <- aov(X~group.factor)
> eta_squared(results)
```

For one-way between subjects designs, partial eta squared is equivalent to eta squared. Returning eta squared.

Effect Size for ANOVA

Parameter	Eta2	95% CI

group.factor	0.20	[0.07, 1.00]

In R

```
> omega_squared(results)
```

For one-way between subjects designs, partial omega squared is equivalent to omega squared. Returning omega squared.

Effect Size for ANOVA

Parameter	Omega2	95% CI
group.factor	0.17	[0.04, 1.00]

Power Analysis in R

```
> # Post-hoc  
> power.f.ancova(eta.squared = 0.2011, factor.levels=5,  
+               k.covariates=0, n.total=100)
```

```
+-----+  
|                POWER CALCULATION                |  
+-----+
```

One-way Analysis of Variance (F-Test)

Hypotheses

H0 (Null Claim) : $\eta^2 = 0$

H1 (Alt. Claim) : $\eta^2 > 0$

Results

Total Sample Size = 100

Type 1 Error (alpha) = 0.050

Type 2 Error (beta) = 0.014

Power Analysis in R

```
> A priori  
> power.f.ancova(eta.squared = 0.2011, factor.levels=5,  
+               k.covariates=0, power=0.80)
```

```
+-----+  
|                SAMPLE SIZE CALCULATION                |  
+-----+
```

One-way Analysis of Variance (F-Test)

Hypotheses

H0 (Null Claim) : $\eta^2 = 0$

H1 (Alt. Claim) : $\eta^2 > 0$

Results

Total Sample Size = 55 <<

Type 1 Error (alpha) = 0.050

Type 2 Error (beta) = 0.176